

MINING GOES GREEN WITH HYDROGEN-FUEL TRUCKS

The pioneering initiative will instill confidence in other industry leaders and can reduce transportation costs.

For a rapidly growing and energy-intensive country like India, the use of hydrogen fuel cell-powered trucks promise to be a major gamechanger. In capital-intensive sectors such as mining and logistics, where diesel dependency is extremely high, these new-age trucks are seen as a harbinger of sustainability.

Data shows that hydrogen-fuel trucks are not only environmentally beneficial but also hold the key to India's net-zero goals by 2070. To be sure, traditional diesel trucks emit large volumes of carbon dioxide, nitrogen oxide, and fine particulate matter (PM2.5), aggravating air and noise pollution in industrial zones.

The matter has come to a head since several Indian cities occupy the unenviable position to be featured among the most polluted in the world, according to IQAir, a Swiss company that tracks global air quality, looking specifically at PM2.5, one of the smallest but most dangerous pollutants.

In this scenario, hydrogen-fuel trucks, which emit only water vapour while operating, have emerged as a clean and sustainable alternative in a massive global battle against climate change. Studies have revealed that an increasing emission of greenhouse gases (GHGs) such as carbon di-oxide, which trap heat in earth's atmosphere, is causing an unprecedented rise in global temperature and triggering extreme weather conditions like floods, droughts, and cyclones.



Chhattisgarh Chief Minister Vishnu Deo Sai presents the key to the nation's first hydrogen-powered truck in Raipur on May 10.

Heavy diesel trucks are also contributing to the pollution mix in a big way. An estimate suggests a single heavy diesel truck emits up to 150 tonnes of carbon dioxide annually. On the contrary, hydrogen trucks are 100% emission-free.

But perhaps, the most cogent argument in favour of the switch to hydrogen trucks is economic.

India spends over $260 billion annually on importing energy sources—primarily diesel and other fossil fuels. Hydrogen-powered trucks can reduce the import burden and boost domestic energy security in a big way.

Hydrogen fuel cell technology also positions India as a global innovation hub. Thanks to initiatives by companies like the Adani Group, India has the opportunity not just to be a consumer of clean technologies, but also a producer and global exporter.

The deployment coincides with over 100 newly-auctioned coal commercial blocks set to begin production. Additionally, the